

The empirical recurrence for OEIS A232051, recovered from its tail

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Abstract

OEIS A232051 records “Empirical recurrence of order 78” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 276$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 85$. Its last coefficient is $c_{78} = -6080 \neq 0$, so no further tail satisfies anything shorter; any order-78 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232051 is “Number of $5 \times n$ 0..1 arrays with no element less than a strict majority of its horizontal and antidiagonal neighbors.”. It has offset 1 and begins

32, 200, 2477, 29974, 377676, 4588174, 55505057, 672197205, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:50:33, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 276$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 276$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 78: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 78.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 552$ exact terms from $n = 7$ on, it returns order 78 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	32	c_{21}	21807868566	c_{41}	5596403614247	c_{61}	−311776184387
c_2	−477	c_{22}	−8938090521	c_{42}	−6160312736654	c_{62}	233521409887
c_3	4828	c_{23}	16044720754	c_{43}	−584425698642	c_{63}	158885167991
c_4	−36601	c_{24}	−105492343541	c_{44}	5981213247288	c_{64}	80319526607
c_5	222109	c_{25}	166020260356	c_{45}	−2512333168192	c_{65}	17839549440
c_6	−1112998	c_{26}	−33756151086	c_{46}	−5877664514838	c_{66}	−34744155776
c_7	4709456	c_{27}	−52027972649	c_{47}	3501855539800	c_{67}	−16440491272
c_8	−16977810	c_{28}	−204496467252	c_{48}	8128131871643	c_{68}	361432162
c_9	52281656	c_{29}	365161945708	c_{49}	−10862437784549	c_{69}	2519020716
c_{10}	−136710704	c_{30}	−21658304942	c_{50}	3853567762764	c_{70}	685962639
c_{11}	300713538	c_{31}	−192005417073	c_{51}	5377344946469	c_{71}	231617735
c_{12}	−554680997	c_{32}	−335546178848	c_{52}	−13235193613269	c_{72}	17508483
c_{13}	892069555	c_{33}	827515904491	c_{53}	7865152360142	c_{73}	−51333687
c_{14}	−1436320886	c_{34}	−19617055693	c_{54}	8474169094475	c_{74}	−18385477
c_{15}	2645463952	c_{35}	−1088872692977	c_{55}	−9099011619175	c_{75}	−49819
c_{16}	−4872485914	c_{36}	−287447457752	c_{56}	−4101766293623	c_{76}	623788
c_{17}	6956741188	c_{37}	2418587156771	c_{57}	3822957817846	c_{77}	20840
c_{18}	−7020223044	c_{38}	139079501312	c_{58}	3930677680723	c_{78}	−6080
c_{19}	7875641368	c_{39}	−4124259648044	c_{59}	−1097549693560		
c_{20}	−15676092966	c_{40}	1201685620776	c_{60}	−1519497073653		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 85$, and it is the recurrence OEIS A232051 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{78} - c_1 x^{77} - \dots - c_{78}$. Its constant term is $-c_{78} = 6080$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 78 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 78, so $\deg q = 78$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 78, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 78, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -6080 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-78 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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